

Siyanda Matika

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Data and AI Lead in Johannesburg, with 8 years of experience across banking, capital markets, insurance and research. I lead data and AI initiatives from scope to production, across the full range of data: large structured datasets in SQL, Spark and Hadoop, ETL pipelines, statistical analysis, and unstructured text, image and video data, through to machine learning and LLM systems in production. In research, I worked on international science experiments in radio astronomy, collaborating on the science and data pipelines that processed petabyte-scale telescope data for the Hydrogen Epoch of Reionization Array (HERA), a precursor experiment to the Square Kilometre Array (SKA).

EXPERIENCE

Manager: AI & Data Science · Discovery Insure

Aug 2024 – present

- Own AI delivery across conversational AI, driver-coaching personalisation and the ML platform behind them. Manage the data science team and run technical hiring for the function.
- Lead the customer virtual agent for WhatsApp and web end to end: scope, agent architecture, retrieval over product documentation, guardrail strategy and integration into customer channels.
- Own the agent-routing classifier, fine-tuned on Azure OpenAI Service to read each user's intent and route the conversation to the right agent.
- Accountable for taking AI systems through privacy and security sign-off: authored the Data Protection Impact Assessments (DPIAs), coordinated multiple penetration tests with the security team, and set up data governance and adversarial testing for chatbots and LLMs.
- Established monitoring for model services on Prometheus and Grafana, with alerts on cost, warnings and errors that automatically trigger pipelines.
- Designed the KubeFlow pod infrastructure for agent development: agents run in isolated Kubernetes pods cut off from the wider network, so an agent that executes tools or code can only reach the services it is explicitly allowed to use.
- Defined the driver-coaching personalisation roadmap, moving from one fixed reward structure for every driver to coaching that adapts to each driver's behaviour.
- Co-led a transformer model on sequential health data with Discovery Health; presented it at the Discovery Actuarial and Data Science Conference, where it placed second.

Machine Learning Engineer · Navigare Securities

May 2022 – Aug 2024

- Joined as the first ML hire and worked at lead level: chose the stack, set the roadmap, and managed the AI product portfolio for capital-markets clients.
- Sat on the hiring panel for the ML team: interviewed and hired three machine learning engineers, then managed and mentored them.
- Owned GPU infrastructure and model deployment: served and fine-tuned open-weight LLMs (Llama, Mistral, Qwen), tuned serving for concurrent users, and introduced RunPod scheduling with run/sleep cycles to keep GPU spend bounded.
- Designed and ran the AWS ML stack, provisioned with Terraform: serverless model services on ECS and Fargate, CI/CD pipelines on GitHub Actions, and data pipelines on Spark and EMR with Delta Lake.
- Led conversational AI for capital-markets clients, including the decision to design an in-house hybrid retrieval framework when off-the-shelf retrieval fell short on precision.
- Conceived and led Elections Navigator, a public voter-education chatbot for South Africa's 2024 national elections, from scope to launch against the fixed election deadline.

Data Scientist · Absa Group

Jan 2020 – May 2022

- Owned the customer lifetime value model and the merchant-device cross-sell recommender from development to production; both were adopted by the marketing and cross-sell teams.
- Delivered customer spend forecasting and customer-health monitoring on Spark and Hadoop over the bank's customer data, and deployed the monitoring dashboards in Docker containers.
- Managed engagement with marketing and cross-sell stakeholders, turning model results into campaign and value-proposition recommendations presented to senior stakeholders.
- Interviewed and hired data scientists for the team, then mentored them and managed their day-to-day work.

Business Analyst · Absa Group

Jan 2017 – Jan 2018

- Owned analyses of retail banking data on trends and customer segments.
- Managed engagement with product, risk and marketing stakeholders, turning the analysis into recommendations for their teams.

SKILLS

Data engineering	Pipelines from raw source to modelling table, at two scales: petabytes of radio-interferometer data in international science collaborations, and bank-scale customer data in Spark, Hadoop and Databricks with Delta Lake for versioned tables. SQL is the working language; Python and Bash do the rest.
Statistics and modelling	Bayesian inference, time-series forecasting, segmentation and recommendation, and A/B tests designed for the power they need and read with the error bars they deserve. Classical machine learning through deep learning in PyTorch, scikit-learn and XGBoost, with evaluation that separates a real gain from single-seed noise.
Language models	Open-weight models (Llama, Mistral, Qwen) and Azure OpenAI in production. Hybrid retrieval built and measured end to end, fine-tuning with LoRA, reinforcement learning with GRPO, intent classification and agent routing, and guardrails tested adversarially before anything reaches a customer.
Unstructured data	Text, images and video: extraction and OCR, embeddings, and retrieval systems evaluated against held-out sets rather than demos.
Infrastructure and deployment	AWS (SageMaker, ECS, Fargate, EMR), Azure (Azure ML, App Service) and GCP, provisioned with Terraform. Kubeflow on Kubernetes with network-isolated pods for agent workloads. GPU serving with vLLM on Linux servers and RunPod, tuned for concurrent users and scheduled to keep spend bounded. Docker, FastAPI and Flask services, CI/CD on GitHub Actions and Azure DevOps.
Operations and monitoring	Prometheus and Grafana for services, model and drift monitoring for what they serve, and alerts wired to pipelines so that cost, warnings and errors trigger an action rather than an email.
Security, identity and governance	IAM on AWS and Azure, Amazon Cognito and App Service authentication. Data Protection Impact Assessments, penetration-test reviews with security teams, adversarial testing of chatbots and language models, and data governance.

EDUCATION

PhD research in Astrophysics, not completed · Rhodes University

2018 – 2019

Research on the Epoch of Reionization with the Hydrogen Epoch of Reionization Array (HERA), a precursor to the Square Kilometre Array (SKA).

- Collaborated on pipelines to process and analyse petabyte-scale radio interferometer data, as part of the international HERA collaboration.
- Applied Bayesian inference and Gaussian process regression to separate the faint cosmological signal from foregrounds.

Co-author of:

- Foreground modelling via Gaussian process regression: an application to HERA data, *Monthly Notices of the Royal Astronomical Society* 495(3), 2813–2826, 2020
- Detection of cosmic structures using the bispectrum phase. II. First results from application to cosmic reionization using the Hydrogen Epoch of Reionization Array, *Physical Review D* 102, 022002, 2020

MSc Astrophysics and Space Science · University of Cape Town and University of the Western Cape

2015 – 2017

Joint UCT–UWC degree through the National Astrophysics and Space Science Programme (NASSP). Thesis: *Detecting Galaxies in the Epoch of Reionization*, supervised by Mario Santos: a Bayesian framework for fitting the 21-cm absorption feature in radio galaxy spectra, to measure the redshift of galaxies in the Epoch of Reionization. Also ran SKA1-LOW noise simulations and large-scale early-universe simulations in C++.

BSc Honours Astrophysics and Space Science · University of Cape Town

2014

BSc Physics · University of the Western Cape

2011 – 2013